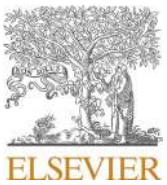
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Original article



How to quantify multidimensional perception of urban parks? Integrating deep learning-based social media data analysis with questionnaire survey methods

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ABSTRACT

Urban parks are places where people regularly connect with nature and each other. Quantifying perceptions of urban parks presents significant changes. Recently, social media data has been increasingly used for studying landscape perceptions, preferences, and management. With the advent of deep learning techniques, the performance of NLP tasks has seen considerable improvement. We posed research questions at the methodological level: How could deep learning-based NLP methods be constructed to assess the multidimensional perception (MDP) of urban parks? How could the assessment performance of this method be validated? In this study, we constructed an MDP of urban parks assessment model based on ERNIE and subsequently conducted a questionnaire survey. By comparing the differences and similarities between the two data sets, we verified the model's assessment performance and proposed the application potential of deep learning-based methods. The findings indicated: (1) our model effectively obtained and assessed sentiment information from online reviews about park accessibility, safety, aesthetics, attractiveness, maintenance, and usability with an accuracy rate exceeding 80 %. (2) The questionnaire survey data confirmed the model's high efficacy, showing consistency in accessibility, aesthetics, and maintenance, but inconsistency in attractiveness and usability due to differences in data expression and timeliness. (3) Deep learning-based NLP methods significantly enhanced sentiment analysis of social media data, showing great potential for practical applications. The results could enhance the performance of sentiment analysis on social media data, serving as a decision-aid tool for park managers and policymakers, and providing valuable insights and guidance for park construction and management.

1. Introduction

Urban parks are places where people daily interact with nature and each other, providing numerous health and well-being benefits for urban dwellers (Hartig et al., 2014; Taylor et al., 2018). These benefits are significantly influenced by the parks' physical characteristics, as well as by visitors' perception and use. Environmental perception represents the process by which people acquire and understand information via sensory perception (Mena et al., 2020). According to the affordance theory, people perceive objects or scenes as encompassing not only information about the environment but also insights into their underlying purpose or motivation (Gibson, 2014). Consequently, more in-depth research is imperative to understand how people perceive the environmental features of parks and to what extent these features influence perceptions of urban parks (Ives et al., 2017). This understanding will enable practitioners to tailor green space policies more effectively in response to public feedback, ultimately better serving the community.

Quantifying perceptions of urban parks is challenging. We seldom utilize our senses in isolation; instead, we experience them in a coordinated, multi-sensory manner (Grahn and Stigsdotter, 2010). Various frameworks have been developed to evaluate environmental characteristics and human preferences. For instance, Kaplan and colleagues established a preference matrix based on human needs and information availability, incorporating four key informational variables to assess environmental features and predict people's preferences. Additionally, the concept of "Perceived Sensory Dimensions" for urban green spaces,

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1618-8667/© 2025 Elsevier GmbH. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

developed by Grahn and Stigsdotter (2010), evaluates the perceived qualities of urban green spaces across eight dimensions. Furthermore, Mikel Subiza-Pérez proposed the "Perceived Environmental Aesthetic Qualities Scale" for evaluating green-blue spaces. Moreover, numerous studies have underscored the significance of physical attributes in shaping landscape perception (Wan et al., 2020; Pot et al., 2021; Luo et al., 2022; Tang et al., 2023; Wei et al., 2023). For instance, greenery, water bodies, and recreational facilities are intricately linked to the perceptions of urban parks (Wang et al., 2017; Jahani and Saffariha, 2020; Jo and Jeon, 2021; Tang et al., 2023; Wei et al., 2023). The frequency with which people perceive various environmental features can reflect their primary concerns, thereby revealing public landscape preferences (Wan et al., 2021; Wei et al., 2023; Ruotolo et al., 2024; Zheng et al., 2024).

In recent years, the use of social media data to study public perceptions and interactions with landscapes has seen an increasing trend (Young et al., 2020; Wilkins et al., 2021; McKitrick et al., 2022; Zabelskyte et al., 2022). Social media data can effectively reflect the environmental messages and behavioral responses that are most directly perceived by the public, offering insights into landscape perception, preferences, and management (Bubalo et al., 2019; Wilkins et al., 2021; Liu et al., 2022; Jiang et al., 2023). Employing social media data to quantify perceptions of urban parks enables a breakthrough in overcoming temporal and spatial limitations, facilitating large-scale online assessments. For instance, studies have shown that public perceptions of environmental messages often include feelings of security, tranquility, and relaxation (Grilli et al., 2021; Polas et al., 2022). Within the textual content, individuals' perspectives, emotions, attitudes, and sentiments towards urban parks can be evaluated through sentiment analysis (Wang et al., 2021b). Sentiment analysis transformed intangible landscape perceptions into numerical scores, facilitating evaluation across a vast array of comment texts (Guo, 2022). This method has been widely used to quantify urban park perceptions (Lin et al., 2022; Grzyb and Kulczyk, 2023; Huai et al., 2023; Qiao and Yeh, 2023; Wei et al., 2023; Yang et al., 2024), offering insights into public needs, attitudes, and emotions, thereby aiding in decision-making and policy formulation.

The latest research methods involved performing sentiment analysis on social media data using Natural Language Processing (NLP) methods (Park et al., 2021; Maiya, 2022). At the methodological level, sentiment analysis can be categorized into supervised and unsupervised approaches. Supervised methods rely on annotated sentiment data, where models learn from this labeled data to recognize patterns of sentiment (Ghahramani et al., 2021; Zhou et al., 2023). For example, previous studies have utilized ready-made sentiment analysis tools (Lin et al., 2022; Li et al., 2024), machine learning methods, and sentiment lexicon-based approach (Wang et al., 2021b; Sun et al., 2023; Zhou et al., 2023) to analyze people's attitudes (Zhou et al., 2023), satisfaction (Wang et al., 2021b), and experiences (Li et al., 2024) of urban parks, correlating perceptions with landscape elements (Huai and Van De Voorde, 2022; Zheng et al., 2024). Unsupervised sentiment analysis does not require pre-annotated sentiment labels; instead, it utilizes clustering (such as k-means clustering) or frequency-based approaches (Wang et al., 2018) to extract sentiment information. With the development of various sentiment analysis tools, such as TextBlob library in Python (Park et al., 2022), Tencent Cloud NLP (Tang et al., 2023), and Baidu NLP (Li et al., 2024), the application of unsupervised methods has become less common.

Previous studies extracting sentiment analysis from urban parks have faced issues of single-dimensionality and inadequate information extraction, with accuracy constrained by existing corpora. For example, sentiment analysis tools can quickly provide judgments of sentiment orientation, but their precision may not match that of customized models. Machine learning methods and sentiment lexicon-based approaches rely on predefined word lists with assigned sentiment values, but they may have limitations in capturing nuanced emotions in complex texts and are limited to analyzing pre-selected text samples. With the advancement of deep learning, deep learning-based NLP can transform unstructured linguistic data into formats recognizable by computers, subsequently facilitating rapid sentiment analysis in online reviews (Young et al., 2018; Lauriola et al., 2022). Compared to other language representation models like ELMo (Embeddings from Language Models) (Peters et al., 2017) and BERT (Bidirectional Encoder Representations from Transformers) (Devlin et al., 2019), ERNIE (Enhanced Representation through Knowledge Integration) (Sun et al., 2019) incorporates more extensive lexical, syntactic, and semantic information from the corpus. This approach enables a more comprehensive capture of latent information in the training corpus and facilitates deeper extraction of text features (Sun et al., 2019). Although deep learning-based sentiment analysis has been applied in other domains (Zhou et al., 2024), there have been no reported applications for quantifying multi-dimensional perceptions of urban parks.

Consider the great potential of deep learning methods for various NLP tasks in terms of social media data analysis. At the methodological level, we posed three research questions: (1) How could the multidimensional perception (MDP) of urban parks be quantified using deep learning-based NLP methods? (2) How could the evaluative efficacy of deep learning-based NLP methods be validated? (3) How could one select a deep learning-based NLP method to maximize its application potential?

Our research design synthesized two distinct types of data. The study aimed to (1) quantify the MDP of urban parks using deep learning-based NLP of social media data, (2) validate the model's assessment efficacy through questionnaire methodology, and (3) explore the application potential of deep learning-based NLP methods. This study pioneered the application of deep learning-based NLP methods in the sentiment analysis of social media data, developing an MDP of urban parks assessment model based on ERNIE capable of automatically quantifying extensive online comment texts into multidimensional sentiment scores. The findings contributed to enhancing the performance of sentiment analysis methods for social media data, expanding the potential of utilizing social media data and deep learning-based NLP methods, and offering practitioners strategic insights and recommendations for park design and management.

2. Methods

2.1. Selection of case parks

Criteria for case parks selection included: (1) parks situated in high-density urban areas, (2) access to comprehensive social media data, (3) capability to conduct on-site questionnaire surveys, and (4) a foundation of prior research (Yang et al., 2021; Cai et al., 2022; Zhou and Lin, 2022).

Four urban parks in Guangzhou were selected: Zhujiang Park, Tianhe Park, Liuhuahu Park, and Renmin Park, with respective areas of 28 hm^2, 70 hm^2, 54.43 hm^2, and 4.46 hm^2. Each of these parks is a comprehensive facility, combining excursion, recreation, and leisure functions. They are publicly accessible, conveniently located for transportation, and managed by dedicated administrative bodies. Design-wise, Zhujiang and Tianhe Park feature a natural mountain-and-water layout, Liuhuahu Park is predominantly organized around its circular lake, and Renmin Park adopts a geometrically symmetrical design. Detailed information can be found in Fig. 1.

2.2. Research design

In this study, we referenced various tools including the Neighborhood Green Space Tool (NGST) (Gidlow et al., 2012), Natural Environment Scoring Tool (NEST) (Gidlow et al., 2018), Bedimo-Rung Assessment Tools (BRAT) (Bedimo-Rung et al., 2006), Environmental Assessment of Public Recreation Spaces (EAPRS) tool (Saelens, 2006), and RECITAL (Knobel et al., 2021), along with literature on park

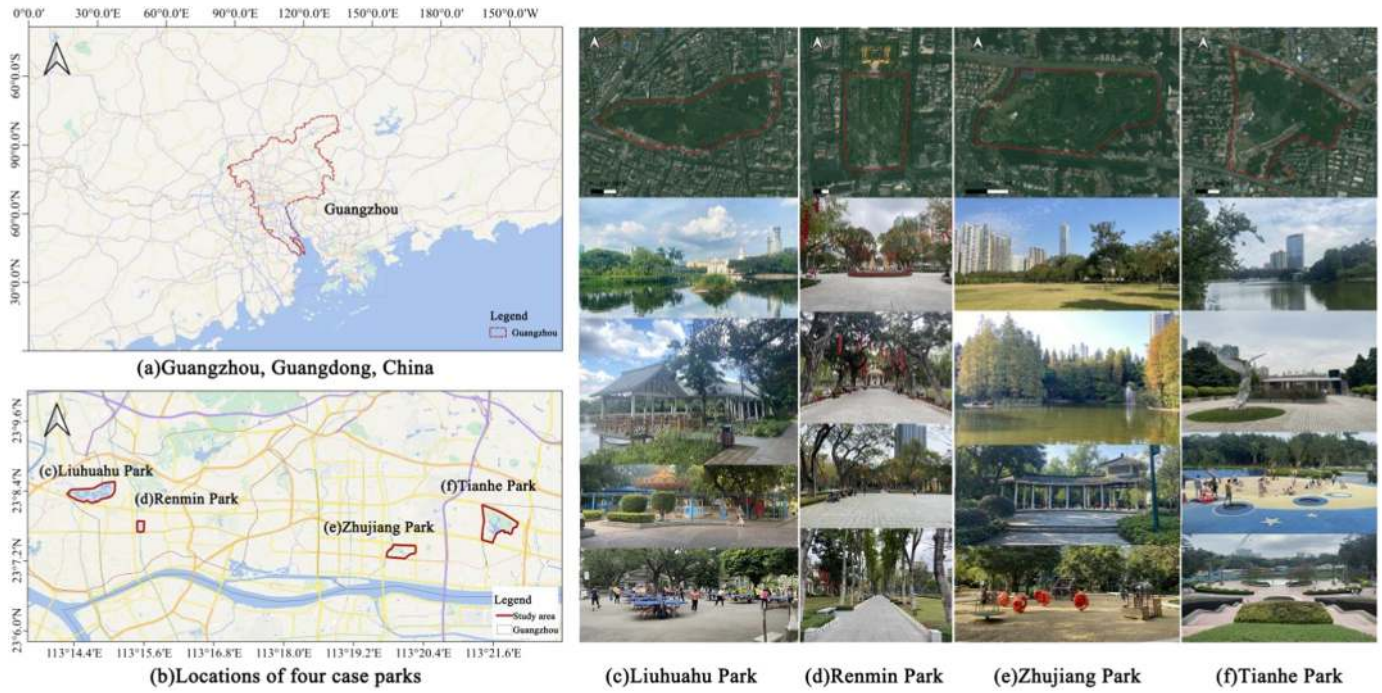


Fig. 1. Locations and photos of four case parks in Guangzhou, China.

perception mentioned in the introduction, to categorize the environmental features of urban parks into six dimensions: accessibility, safety, aesthetics, attractiveness, maintenance, and usability. The selection of these dimensions considered the availability of data, the independence between dimensions, and their widespread acceptance in both academic and practical domains, ensuring the scientific rigor, practicality, and broad applicability of our assessment framework.

In this study, we employed a sequential mixed-methods design within the framework of mixed research methods, which was divided into three stages as depicted in Fig. 2. In stage 1, we utilized comments from Dianping (http://www.dianping.com/) as our primary social media data, which was particularly suitable for analyzing perceptions of urban parks (Wang et al., 2021a).

Stage 1: Constructed an MDP of urban parks assessment model based on ERNIE using social media data from Dianping. This model quantified the MDP of urban parks and extracted keywords for each dimension from the comments. This stage comprised five steps: (1) Utilized Python crawler to collect user comments on parks from Dianping, followed by screening and cleaning processes. (2) Randomly select a subset of comments, and manually label them across six dimensions to create a training set. Train the ERNIE model using this data, and choose the parameters that yield the best classification results as the optimal settings. (3) Used the trained model to assess new comments to obtain both positive and negative evaluations of the MDP of urban parks within these texts. (4) Generated a co-occurrence matrix from positive and negative comments using Python. (5) Imported the co-occurrence matrices for each dimension into Gephi for data visualization.

Stage 2: Analysis of respondents' perceptions of urban parks using questionnaire data. This stage involved preparing a questionnaire, informed by the results of Stage 1, to gauge the MDP of urban parks' environmental features. The questionnaire was disseminated and collected through field surveys, enabling the analysis of respondents' evaluations regarding the MDP of these urban parks.

Stage 3: Data analysis and integration. This stage involved comparing the results of social media and questionnaire data analyses and utilizing the latter's results to validate the efficacy of the newly proposed model in this study.

2.3. Deep learning-based NLP methods

The stage of constructing the MDP of urban parks assessment model based on ERNIE, depicted in Fig. 3, involved data collection, model training, and data visualization.

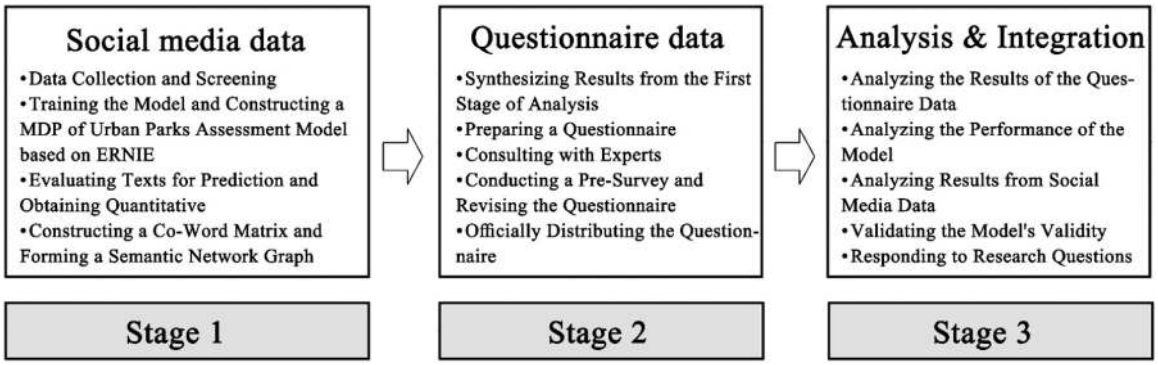


Fig. 2. Sequential mixed-methods design.

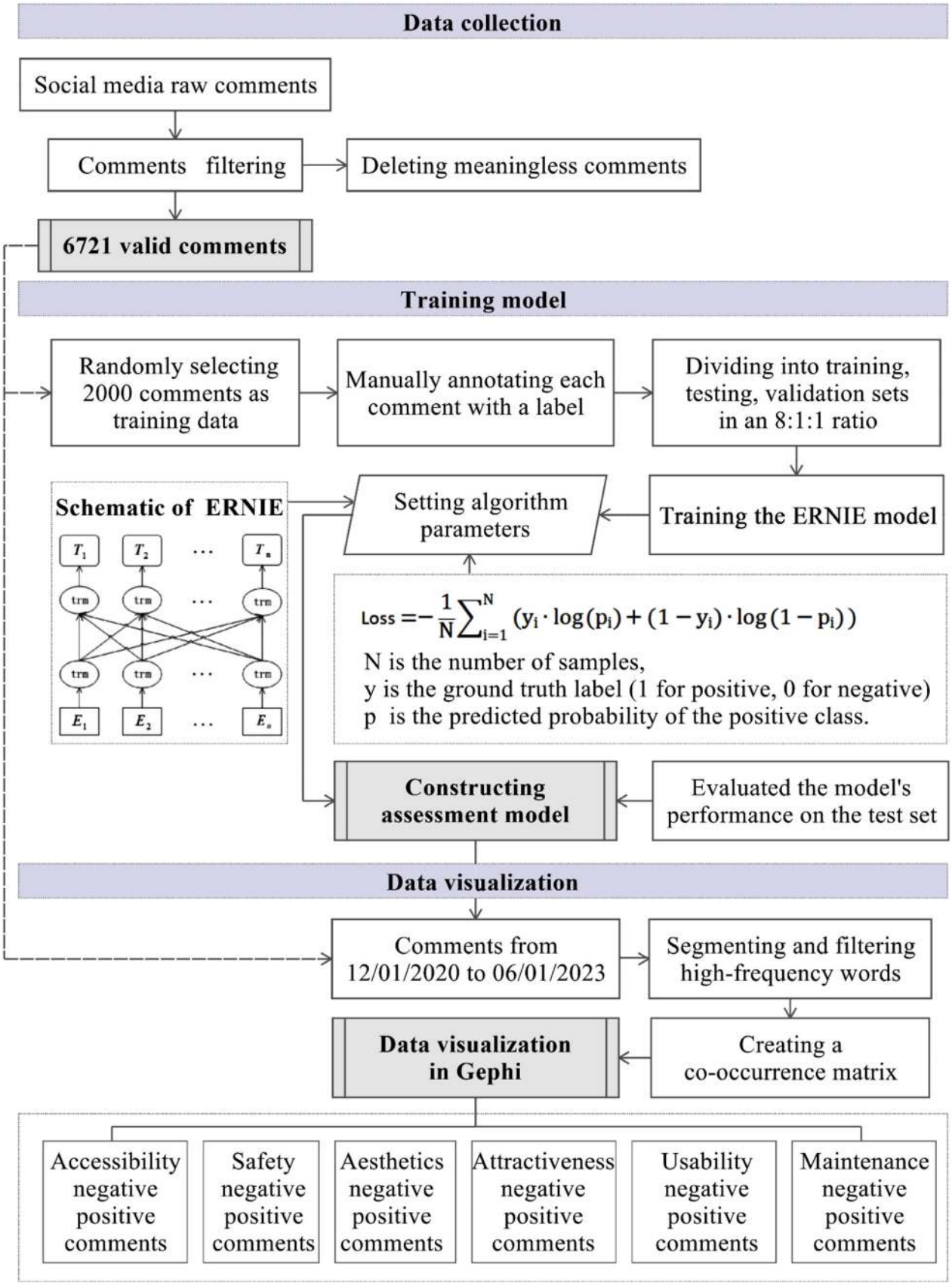


Fig. 3. Modeling the MDP of urban parks assessment model using deep learning-based NLP methods.

2.3.1. Social media data collection

Using Dianping as the source of social media data, data were collected from four urban parks, namely Zhujiang Park, Tianhe Park, Liuhuahu Park, and Renmin Park. Based on the historical context of urban park development in Guangzhou and pivotal events in the city's epidemic prevention and control, the mandate for wearing masks in public places was officially enforced on January 26, 2020. Subsequently, from November 30, 2022, Guangzhou systematically relaxed control measures across all districts. In the context of the epidemic era, urban park renovation activities experienced a notable slowdown, yet the overall status of the parks remained consistently maintained. Therefore, this paper collects comments on the four parks from the establishment of Dianping in April 2003 to June 1, 2023.

Firstly, we employed Python for web scraping to collect data, resulting in 6768 comments, of which 6721 were deemed valid. Each comment included park names, user IDs, ratings, comment texts, and timestamps. Secondly, we randomly selected 2000 comments before December 2020 and annotated the data from six dimensions to create a training dataset. This dataset was split into training, testing, and validation sets in an 8:1:1 ratio. Utilizing this dataset, we trained the ERNIE model and selected the parameters that yielded the best classification performance as the optimal parameters for the algorithm. Lastly, we used the trained model to assess park comments from December 1, 2020, to June 1, 2023, totaling 3388, and utilized the pandas Python Library to generate a co-occurrence matrix from the evaluation results. This matrix was visualized in Gephi to facilitate comparative analysis between subsequent questionnaire survey results and the model's evaluation.

2.3.2. Training the ERNIE model

ERNIE, developed by Baidu, is an advanced language representation model that improves upon the BERT model. It uses a knowledge masking strategy and consists of two stages: pre-training and fine-tuning for specific tasks. Unlike BERT, which breaks down word relationships indiscriminately, ERNIE intelligently masks entities, words, and phrases, learning correspondences at various levels. This allows ERNIE to simultaneously utilize lexical, syntactic, and semantic information, significantly enhancing its performance, particularly in its masking mechanism (Devlin et al., 2019; Sun et al., 2019).

The specific steps of model training are as follows:

(1) For the dataset established in Section 2.3.1, we constructed an urban park perception dataset by manually annotating each comment (ground truth) with a label from the six established dimensions. For example, regarding the "accessibility" dimension, we used "0" to indicate a negative sentiment reflecting poor accessibility in the comment, "1" to indicate positive accessibility, and "2" to indicate that the comment does not reflect any information about accessibility. The same labeling scheme was applied to the remaining five indicators, depicted in Fig. 4. In this process, the first and second authors were responsible for manual annotation, while the corresponding author verified and made final decisions.

(2) We initially pre-trained the ERNIE model on the large-scale Chinese language corpus, which incorporated diverse data sources such as encyclopedia, news, dialogue, information retrieval, and discourse relation data from a search engine (Sun et al., 2019). This extensive dataset ensured that ERNIE captures a comprehensive understanding of the Chinese language across different genres and registers.

(3) We fine-tuned the model on our curated dataset to enhance its performance for text classification tasks for each of the six dimensions. The cross-entropy loss function was utilized to compute the loss, represented by the formula:

$$\mathbf{Loss} = -\frac{\mathbf{1}}{\mathbf{N}}\sum_{i=1}^{N}(\mathbf{y_i} \bullet \mathbf{log}(\mathbf{p_i}) + (\mathbf{1} - \mathbf{y_i}) \bullet \mathbf{log}(\mathbf{1} - \mathbf{p_i})) \quad (1)$$

where N is the number of samples, y_i is the ground truth label (1 for positive, 0 for negative), and p_i is the predicted probability of the positive class.

To obtain model predictions, we employed a fully connected layer, and the computation formula is:

$$\mathbf{Prediction} = \boldsymbol{\sigma}(\mathbf{Wx} + \mathbf{b}) \quad (2)$$

where W is the weight matrix, x is the input vector, b is the bias term, and σ is the sigmoid activation function that squashes the output to a probability range between 0 and 1.

We evaluated the model's performance on the test set using four metrics, including Accuracy (ACC), Precision, and F1-score. The formulas are as follows:

$$\mathbf{ACC} = \frac{\mathbf{TP} + \mathbf{TN}}{\mathbf{TP} + \mathbf{TN} + \mathbf{FP} + \mathbf{FN}} \quad (3)$$

$$\mathbf{Precision} = \frac{\mathbf{TP}}{\mathbf{TP} + \mathbf{FP}} \quad (4)$$

$$\mathbf{F1Score} = \mathbf{2} \times \frac{\mathbf{Precision} \times \mathbf{Recall}}{\mathbf{TP} + \mathbf{FN}} \quad (5)$$

where TP is the number of true positives, TN is the number of true negatives, FP is the number of false positives, FN is the number of false negatives.

To showcase the superiority of the ERNIE model, we conducted comparative experiments. Notably, BERT, renowned for its deep bidirectional architecture and rooted in the Transformer architecture, has been employed in sentiment analysis of the built environment by Rui (2023) and He et al. (2023). We fine-tuned both the ERNIE and BERT models for text classification tasks. Specially, the batch size was set to 10, the pad size to 150, the learning rate to 5e-5, and the hidden size to

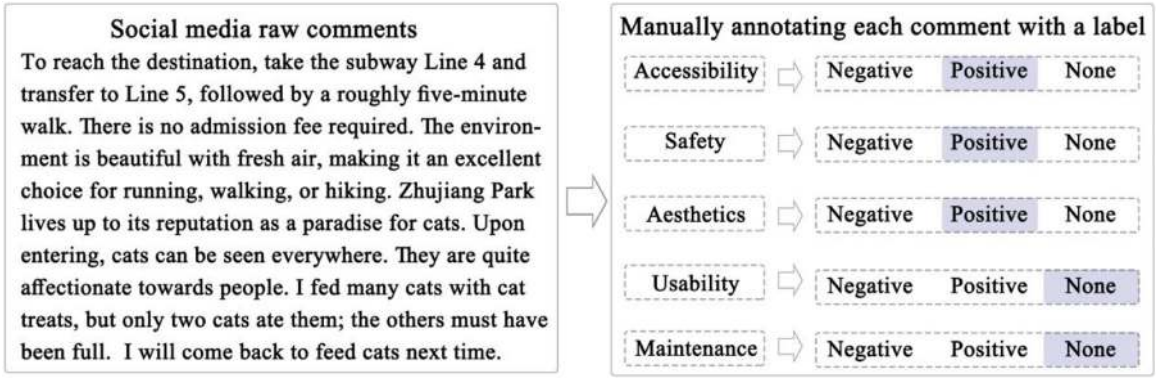


Fig. 4. Manually annotating each comment (ground truth) with a label.

768. An early stopping strategy was incorporated, terminating the training process if the validation set loss showed no improvement over 200 consecutive batches.

We also included classical machine learning algorithms such as XGBoost (Wang, Zhu et al., 2021b), Support Vector Machine (SVM) (Ahmad, Munir, et al.), Long Short Term Memory networks (LSTM) (Li, Fu et al. 2023) as benchmarks, given their prominence in sentiment analysis tasks. We fine-tuned these models using the same dataset developed in Section 2.3.1 for text classification tasks.

2.3.3. Text keyword extraction based on co-occurrence network

To compare results from social media data analysis with subsequent questionnaire surveys, we used a trained model to evaluate park comments from December 1, 2020, to June 1, 2023, generating perceptual scores for six dimensions. This time frame was selected due to the stability of park environmental characteristics amidst pandemic impacts on park construction. Subsequent questionnaire content was designed to align with current park construction and environmental conditions.

We employed the jieba library in Python for text segmentation, using a custom stopword list derived from Harbin Institute of Technology and Baidu to filter out irrelevant words and replace synonyms with unified terms. Using the pandas library in Python, we extracted high-frequency keywords from comments across dimensions to build a co-occurrence matrix.

Finally, we visualized these matrices in Gephi, presenting perceptual scores and key comment information intuitively. A co-occurrence network was constructed, with words as nodes and edges indicating co-occurrence frequency or weight. The PageRank algorithm was applied to identify important or clustered keywords, facilitating the extraction of representative theme-related keywords and illustrating word relationships. In the keyword network graph, node size indicates keyword frequency and connection color darkness shows edge weight. This method helped create an environmental features dictionary.

2.4. Questionnaire survey methods

2.4.1. Questionnaire design and data collection

A questionnaire was designed to evaluate the MDP of urban parks. This was based on findings from the first phase of our study and the assessment tool for urban park environmental features listed in Table 1. The questionnaire was divided into two sections: (1) the personal information of the respondents, which included gender, age, occupation, education level, and annual disposable income; and (2) the respondents' evaluations of the MDP in these urban parks. Participants were asked, "To what extent do you agree with the following descriptions regarding the environmental features of the parks you regularly visit?" They were instructed to reflect on their experiences over the past six months and rate them using a 5-point Likert scale. More details about the scale are shown in *Appendix A*.

The data collection process involved conducting an on-site survey, which included the random distribution and collection of questionnaires. Preliminary surveys were carried out before the formal distribution. The official distribution of the questionnaires took place from August 28 to September 17, 2023. This survey was conducted in various areas of four different parks. Respondents in these areas were randomly selected to complete the questionnaires.

Table 1
Rationale for categorizing the environmental features of urban parks.

Dimensions	Tools for categorizing the environmental features of urban parks
Accessibility	NGST, BRAT, EAPRS, RECITAL
Safety	BRAT, RECITAL
Aesthetics	NGST, BRAT-DO, EAPRS, RECITAL
Attractiveness	BRAT, RECITAL
Usability	NGST, RECITAL
Maintenance	NGSTBRAT, EAPRS, RECITAL

2.4.2. Statistical analysis

Frequency analysis was employed to ascertain the socio-demographic characteristics of the respondents. Descriptive statistics were utilized to outline the basic characteristics of the data. These socio-demographic details aimed to reflect the survey's alignment with the actual situation of parks in Guangzhou. To avoid bias and stereotypes, gender results were not reported. The reliability of the questionnaire was assessed using Cronbach's alpha coefficient (Cronbach, 1951), with a total scale reliability of 0.8 or above and a subscale reliability of 0.7 or above considered good (Landis and Koch, 1977). The questionnaire was used to validate the trained model, so scale validity was not discussed.

3. Results

3.1. Results of the MDP of urban parks based on ERNIE

3.1.1. Model performance and quantitative results

As shown in Table 2, the ERNIE model proved to be the most effective, achieving an average ACC of 86.73 %, which is 0.934 % higher than BERT. It exhibited substantial improvements over the BERT model, with an average precision surpassing 0.0874, and F1 score above 0.1213. These metrics underscore the efficiency of the ERNIE model in capturing perceptions from the park comments.

Using the fine-tuned ERNIE model with optimized parameters, we scored the comments on four parks from Dianping spanning from January 1, 2020 to June 1, 2023. The results provide positive and negative evaluations for six dimensions of environmental features in each of the four parks, as depicted in Table 3.

3.1.2. Keyword extraction based on co-occurrence network

Using co-occurrence network analysis, we extracted keywords from comments related to six dimensions, depicted in Fig. 5 and Appendix B. Our analysis focused on clustering units for each dimension, high-frequency words in each unit, and similarities and differences among high-frequency words across dimensions.

Firstly, we identified high-frequency word clusters in each dimension, comprising 3–5 units. Common units across dimensions included activity types, recreation, and service facilities. For "Accessibility," clusters included recreation and service facilities, and transportation modes. "Safety" had clusters of plants, activity types, personal characteristics, transportation modes, and recreation/service facilities. "Aesthetics," "Attractiveness," and "Usability" included clusters of plants, animals, activity types, personal characteristics, and recreation/service facilities. "Maintenance" clusters included activity types, recreation/service facilities, construction, management measures, and epidemic prevention.

Secondly, we categorized high-frequency keywords into ten groups: (1) Activity types (walking, running, playing), (2) Transportation modes (subway, bus, parking space), (3) Personal characteristics (children, the elderly), (4) Recreation facilities (playgrounds, slides), (5) Service facilities (toilets, convenience stores), (6) Animals (cats, swans), (7) Plant descriptions (bald cypresses, scenery), (8) Construction and development (construction, projects), (9) Management measures (cleaners, gardeners), and (10) Epidemic prevention and control (epidemic, masks).

Thirdly, the analysis showed both similarities and differences in high-frequency keywords across dimensions. Common keywords were walking, free, children, subway, transportation, facilities, and weather. "Aesthetics" and "Attractiveness" overlapped in plant-related terms, with aesthetics focusing on plant beauty and attractiveness including facilities and small animals. "Usability" and "Maintenance" shared keywords on epidemic prevention and construction. Maintenance received negative evaluations due to the impact of epidemic measures and subway construction.

Table 2
Performance of the two text classification models.

Metrics	Models	Accessibility	Safety	Aesthetics	Attractiveness	Usability	Maintenance
ACC	ERNIE	**0.8986**	**0.9808**	**0.8514**	**0.8278**	**0.8026**	**0.8452**
	BERT	0.8311	0.9038	0.7770	0.7483	0.5789	0.8065
	SVM	0.7075	0.9150	0.6599	0.6333	0.5541	0.8377
	XGBoost	0.6190	0.9020	0.5714	0.6067	0.4797	0.7882
	LSTM	0.6599	0.6137	0.6721	0.6200	0.5473	0.7987
Precision	ERNIE	**0.8983**	**0.9812**	**0.8587**	**0.8269**	**0.8023**	0.8338
	BERT	0.8328	0.9131	0.7867	0.7446	0.5560	**0.8439**
	SVM	0.7545	0.9158	0.7587	0.6163	0.6155	0.8421
	XGBoost	0.7104	0.8135	0.6214	0.3838	0.3929	0.7478
	LSTM	0.7664	0.6127	0.6770	0.5051	0.6225	0.7930
F1 score	ERNIE	**0.8977**	**0.9772**	**0.8494**	**0.8273**	**0.7994**	**0.8372**
	BERT	0.8315	0.8582	0.7712	0.7408	0.5386	0.7200
	SVM	0.6590	0.8843	0.6119	0.5037	0.4979	0.7880
	XGBoost	0.5438	0.8555	0.4749	0.4702	0.3565	0.7162
	LSTM	0.6082	0.6759	0.6747	0.5000	0.4496	0.7625

Table 3
Quantitative results of public perception in six dimensions of urban parks.

	Accessibility	Safety	Aesthetics	Attractiveness	Usability	Maintenance
Negative	108	0	41	159	126	268
Positive	1034	262	1427	2571	1875	0
None	2246	3126	1920	658	1387	3120

3.2. Results of the questionnaire

3.2.1. Demographic characteristics

A total of 1416 questionnaires were distributed, and 1318 valid responses were obtained after screening, yielding an effective response rate of 93 %.

The demographic characteristics of the respondents are illustrated in Fig. 6. Most respondents (46 %) were aged between 19 and 35 years. The largest proportion of respondents, 57 %, held a bachelor's or college degree. Employees currently working represented the largest segment of respondents at 40 %. Respondents with high and low incomes were the largest income groups, constituting 31 % and 29 %, respectively. Overall, this study achieved a desirable representation of demographic characteristics, consistent with the actual demographic profile of park users in Guangzhou.

3.2.2. Reliability of the scale

The 1318 samples collected after multiple rounds of screening were all considered valid. The results of the reliability test for the scale are presented in Table 4. The indicator-corrected item-to-total correlations were not found to be excessively low. The calculations indicated that the overall reliability of the scale did not significantly improve upon the deletion of any items.

3.2.3. Analysis of respondents' perceptions in six dimensions of urban parks

Frequency analyses were conducted to assess the perceived level of environmental features across the six dimensions of urban parks, as rated by the respondents. These analyses considered the proportion of individuals who agreed (including both agreement and strong agreement) to rank each measurement indicator from highest to lowest. The overall assessment by respondents of the perceived levels of these measurement indicators across the park's six dimensions is depicted in Fig. 7. As shown in Fig. 7, in the dimensions of aesthetics, attractiveness, and maintenance, recreational and service facilities did not receive as high positive evaluations as plants and water bodies did. Respondents also gave lower ratings for the park's lighting, tranquility, and historical culture.

Compared to social media data, which often provided a unidirectional perspective focusing on aspects like security and maintenance, questionnaire data enabled a more comprehensive evaluation. For each dimension and measurement, respondents showed varying degrees of agreement or disagreement. Notably, the proportion of respondents in agreement exceeded those in disagreement. This aligns with the analysis of social media data, which also indicates a higher frequency of positive evaluations over negative ones.

4. Discussion

4.1. Performance of MDP of urban parks assessment model based on ERNIE

In this study, we employed deep learning-based NLP methods for the first time to analyze social media data sentiment and constructed an MDP of urban parks assessment model based on ERNIE. Our model effectively processed the MDP of parks from numerous comments, determining positive or negative evaluations across six dimensions with over 80 % accuracy. We discovered that high-frequency words in our study formed clusters related to activities, transportation, epidemic control, construction, and facilities, with physical features mainly as nouns and sensory descriptions as adjectives. Negative evaluations often concerned construction, noise, odor, and management, while natural elements and facilities received mixed reviews. We observed similarities and differences in frequent keywords across dimensions, with some overlap in terms like aesthetics, attractiveness, usability, and maintenance. Respondents objectively commented on various park features, indicating that negative aspects did not overly influence overall perception.

Previous research methods for quantifying park perceptions faced three main limitations. Firstly, machine learning-based sentiment analysis, like in studies by Huai and Van De Voorde (2022), was restricted to one-dimensional analysis on specific online reviews, which could contain mixed sentiments. Secondly, this approach required manual annotation, as seen in Wang et al. (2021a), where experts manually categorized 500 text annotations. Finally, machine learning-based NLP, used by Koblet and Purves (2020), didn't assume dimensionality and was limited to the analysis of selected texts, requiring repetition if the texts changed.

Our approach outperformed previous methods in sentiment analysis of social media data by using an advanced, deep learning-based NLP method. Firstly, this method eliminated the need for granular text

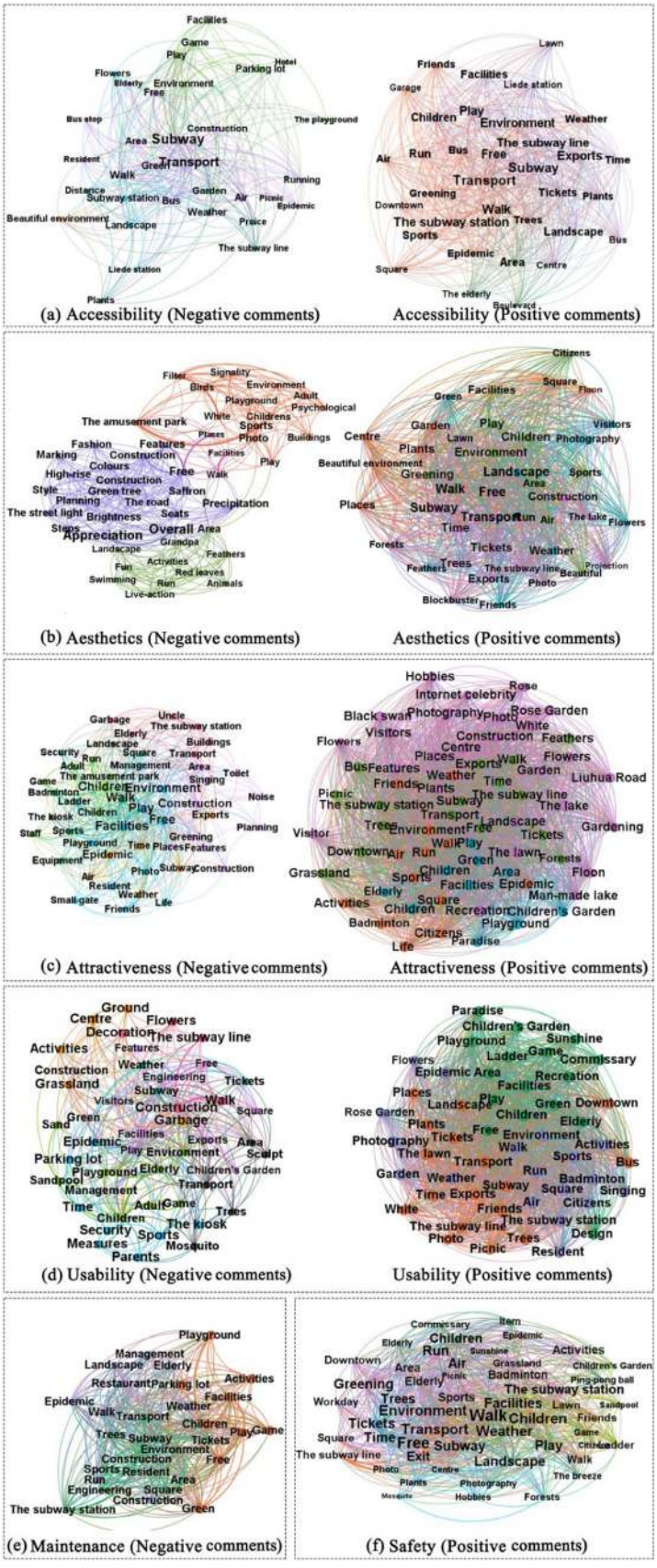


Fig. 5. Co-occurrence network graph of high-frequency keywords.

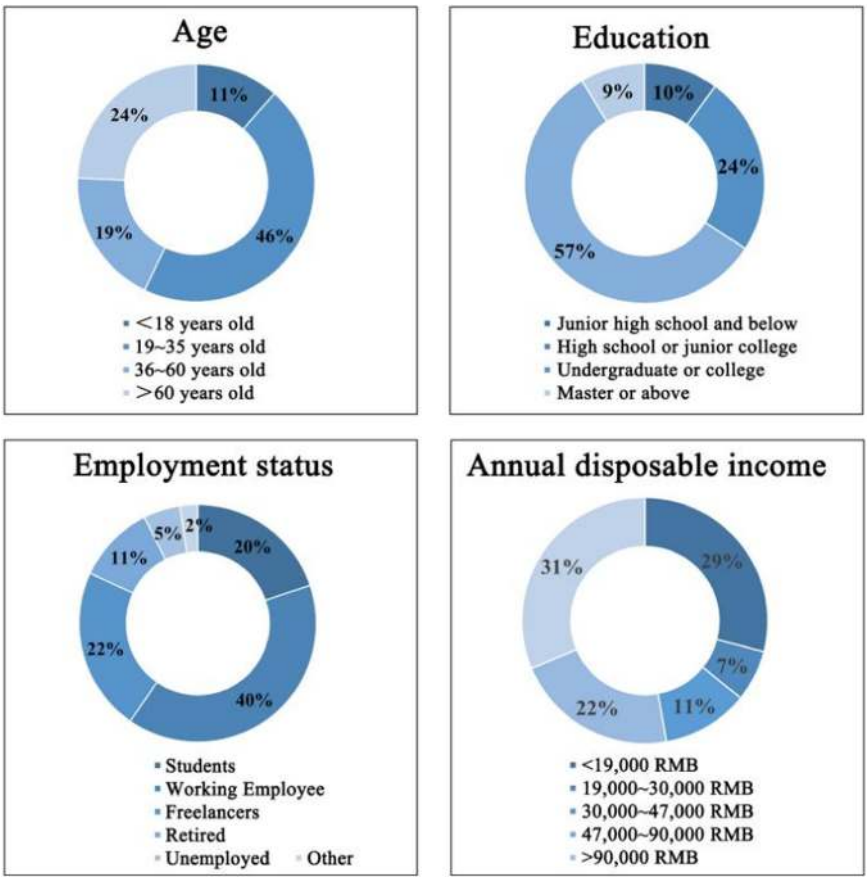


Fig. 6. Demographic characteristics of the respondents.

processing, enabling comprehensive, multidimensional sentiment evaluation of comment paragraphs. Secondly, the deep learning methods required a predefined dimensionality framework and manual text annotation for effective model training. For example, text with both positive comments on aesthetics and negative comments on maintenance needed manual classification for each aspect. Thirdly, our study trained computers to interpret human language, allowing autonomous sentiment analysis of online reviews. This "black box" capability enabled the model to analyze sentiments, positive or negative, in texts of any length, demonstrating its broad applicability.

4.2. Validity of the newly developed model through the questionnaire survey

Comparing the questionnaire and social media data, we found similarities and differences in park perceptions. Both sources consistently evaluated parks on accessibility, aesthetics, and maintenance, with mixed attitudes. Respondents praised parks' accessibility, especially near metro and bus stops, and favored natural elements but rated the "quiet spatial atmosphere" low. The questionnaire provided detailed feedback, such as lower scores for recreational facilities' aesthetics due to varied user interests. Safety was positively rated in both datasets, reflecting effective law enforcement in Guangzhou city parks. However, perceptions of attractiveness and cultural aspects varied among groups. Social media data mostly had negative comments about park maintenance, while questionnaire responses were more mixed. This discrepancy could be due to social media users focusing on maintenance issues, whereas questionnaires covered various maintenance aspects more comprehensively.

There were two possible reasons for the differences in perceptions of these urban parks. Firstly, the manner of data expression: social media data included random comments, while survey data consisted of structured questions and options. Secondly, data timeliness: social media data, due to its large scale and real-time nature, may contain richer information but also noise and biases. Therefore, survey data is more precise and controllable, suitable for small-scale studies or pre-surveys. Social media data is easier to obtain for large-scale surveys. The best approach, when funds allow, is to use both social media and survey data for comprehensive and accurate park perception information. Cross-validation ensures the accuracy and reliability of the data.

Upon synthesizing the two data analyses, we proposed four key points for urban park construction and renewal. Firstly, natural elements were crucial, receiving positive public evaluations. Incorporating plants and water bodies can enhance park attractiveness and positive perceptions. Secondly, increasing recreational and service facilities, such as seats, pavilions, and lights. Regardless of age, the usability of facilities forms the foundation for all activities, emphasizing the importance of convenient man-made amenities (Dallimer et al., 2014, Wang et al., 2021a). Thirdly, the evaluation of noise and lighting by individuals is relatively low, underscoring the critical importance of park management in these areas. Fourthly, effective park construction and renewal require field surveys and diverse opinions. Balancing these views ensures well-rounded, comprehensive analyses without overemphasizing specific groups' perspectives.

4.3. Application potential of deep learning-based NLP methods

The advent of deep learning methods has significantly enhanced sentiment analysis performance in social media data, effectively capturing users' meanings and intentions beyond simple keyword-based filtering. Deep learning-based methodologies model and represent text using multi-layer neural networks, learning abstract and high-level

Table 4
Reliability test for the scale assessing the MDP of urban parks.

Dimensions	Items	Cronbach's Alpha	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
Accessibility	acc1	0.893	0.571	0.907
	acc2		0.772	0.864
	acc3		0.784	0.864
	acc4		0.753	0.868
	acc5		0.729	0.872
	acc6		0.757	0.869
Safety	saf1	0.849	0.692	0.809
	saf2		0.73	0.79
	saf3		0.74	0.785
	saf4		0.605	0.847
Aesthetics	aes1	0.9	0.742	0.883
	aes2		0.718	0.885
	aes3		0.622	0.901
	aes4		0.781	0.879
	aes5		0.744	0.882
	aes6		0.719	0.884
	aes7		0.703	0.886
Attractiveness	att1	0.742	0.454	0.714
	att2		0.607	0.683
	att3		0.546	0.686
	att4		0.427	0.73
	att5		0.39	0.731
	att6		0.54	0.691
Maintenance	mai1	0.911	0.803	0.887
	mai2		0.802	0.886
	mai3		0.734	0.9
	mai4		0.788	0.889
	mai5		0.763	0.895
Usability	usa1	0.878	0.622	0.873
	usa2		0.663	0.865
	usa3		0.767	0.84
	usa4		0.773	0.838
	usa5		0.741	0.845

features that augment the performance and effectiveness of various NLP tasks. In social media data analysis, deep learning methods can further enhance performance by training on domain-specific text data. For instance, models can undergo fine-tuning with labeled data from social media, enhancing their adaptability to the specific features and language styles of social media. Notably, language representation models such as ELMo (Peters et al., 2017), BERT (Devlin et al., 2019), and ERNIE (Sun et al., 2019), through their pre-training on extensive text data, effectively capture the semantic and contextual information of the text.

Compared with previous research, deep learning-based NLP methods offer several advantages. Firstly, they are not limited to specific text types and enable multidimensional sentiment analysis on any form of online reviews. This allows for the inclusion of reviews from various sources such as social media, travel websites, and other online platforms, thereby providing a more comprehensive database. Secondly, these methods eliminate the need for laborious sentence and word segmentation in long texts, allowing for the direct processing of entire paragraphs for multidimensional sentiment analysis. Thirdly, the ability to analyze online comments over various time and spatial scales aids researchers in investigating landscape changes from a longitudinal perspective. Fourthly, researchers can tailor models, selecting features and algorithms as needed for specific applications. Deep learning methods, with their use of labeled datasets, yield results that are more reproducible and facilitate validation by other researchers. However, deep learning methods still require pre-training on extensive datasets with manual annotations, which necessitates a significant investment of time and human resources by researchers in labeling and organizing datasets to ensure the accuracy and reliability of deep learning models.

In practical applications, the model developed in this paper can serve as a decision-making aid for park managers and policymakers. Through sentiment analysis, it provides insights into public satisfaction with the six environmental dimensions of parks, enabling targeted improvements in facilities, services, and management to enhance park quality and appeal. The model quantifies multidimensional park perceptions, including safety, accessibility, aesthetics, attractiveness, maintenance, and usability, assisting practitioners in improving urban parks with low perceived value. The co-occurrence matrix of high-frequency words for environmental characteristics reveals visitor priorities in parks, aiding in defining management focus and priorities.

4.4. Limitations and prospects

There were three limitations in this study. Firstly, the study primarily utilized Dianping as the social media data source, which potentially limited the comprehensiveness of the sample. Secondly, the timeframe for the online reviews analyzed spanned the last three years (2020.12.01–2023.06.01), leading to potential gaps in perceptual information within the reviews. For instance, there was a noticeable absence of positive comments regarding maintenance. Thirdly, the study was confined to the analysis of four parks in Guangzhou, with the model parameters tailored to the Chinese cultural context. For other languages, it necessitated retraining of the model and adjustment of parameters.

We recommended five approaches for future research using social media data. Firstly, research utilizing social media data should comprehensively consider the diversity of park users and the demographics of social media commenters. It is crucial for researchers to accurately grasp the context and limitations of the data to avoid biased conclusions. Secondly, researchers should choose appropriate research methods that align with the study's objectives. Traditional surveys, for instance, are effective for providing detailed contextual insights and understanding respondent traits, behaviors, and opinions. Integrating these with deep learning-based NLP can enhance research comprehensiveness and dimensionality, aiding in the development of more effective design and management strategies. Thirdly, while deep learning-based NLP methods support decision-making, their practical application in design strategies and graphic language requires the expertise of experienced researchers. Researchers should interpret the model's output, extract key insights, and translate them into a clear and actionable format. Fourthly, significant differences exist in the evaluation criteria, expressions, and emotional tendencies of parks across various languages and cultures. Therefore, developing a sentiment analysis model suitable for other languages and cultural contexts entails re-collecting and labeling the relevant corpus and adjusting the corresponding parameters. Fifth, in utilizing online reviews for research, it is imperative to respect individual privacy and eschew unethical behavior.

5. Conclusion

We utilized deep learning-based NLP methods to analyze the sentiment of social media data and developed an MDP of urban parks assessment model based on ERNIE. We verified the validity of the model through questionnaire surveys and proposed the potential application of deep learning-based NLP methods. Our findings included: (1) our model effectively captured and evaluated the positive and negative sentiment information of online reviews regarding park accessibility, safety, aesthetics, attractiveness, maintenance, and usability. (2) The questionnaire data confirmed the strong validity of the model. The negative elements of the parks were predominantly artificial elements, while the park's natural elements received positive feedback. However, there were notable similarities and differences in the park perception information reflected by the questionnaire data and social media data. The two types of data showed consistency in terms of accessibility, aesthetics, and safety, but inconsistency in terms of attractiveness and maintenance. (3) The deep learning-based NLP methods were capable of significantly enhancing the performance of sentiment analysis of social media data, demonstrating substantial potential for application due to their unique advantages.

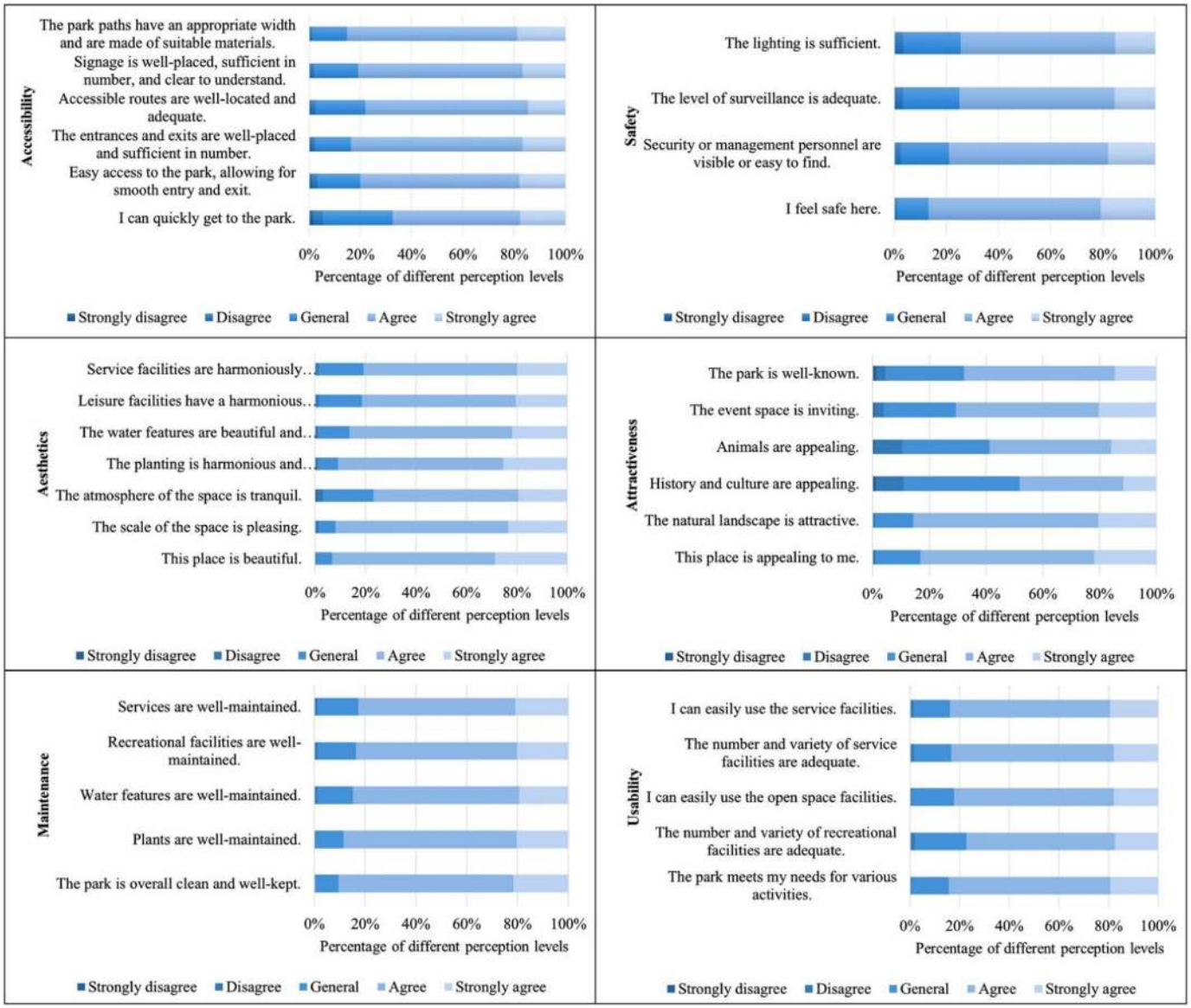


Fig. 7. Distribution of respondents' perceptions across the six dimensions of urban parks.

Compared to previous studies, the model developed in this paper exhibited superior performance in sentiment analysis and facilitated automatic assessment of multidimensional positive or negative sentiment for many texts of arbitrary length, highlighting its significant contribution to sentiment analysis of social media data and deep learning-based NLP methods. This model also served as a decision-making tool to assist park managers and policymakers. By analyzing the sentiment tendencies of users on social media, a deeper understanding of the public's opinions and demands for parks was attained, thus enabling optimization of park construction and management. The results of the study positively impacted the enhancement of sentiment analysis performance of social media data, advanced the application potential of deep learning-based NLP methods, and offered valuable insights and references for park construction and management.

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CRediT authorship contribution statement

Huang Wenwen: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Zhao Xukai**: Writing – original draft, Methodology, Formal analysis. **Lin Guangsi**: Writing – review & editing, Supervision, Methodology, Funding acquisition. **Wang Zhifang**: Writing – review & editing. **Chen Mengyun**: Writing – review & editing..

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

A. ppendices A and B: Supporting Information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ufug.2025.128754.

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